

Testicular germ cell tumor

Most common solid malignancy in males 15–35 · **>90% cure, >95% 5-yr survival** — chemotherapy is curative even in advanced disease · lung is the commonest met site (62%); liver (16%) and brain (7%) are negative prognostic factors

Stanford Oncology Handbook 2025, Testicular Cancer (Moore K, Glover M, Shah S) · NCCN v1.2025

WORKUP

- Suspicious mass: H&P, **AFP, bHCG, LDH**, chemistry, **testicular ultrasound**
- **Repeat all markers after orchiectomy** — TNM staging uses post-orchiectomy values
- **Pure seminoma** (seminoma histology + **normal AFP**; bHCG may be elevated): abdominal/pelvic CT + **chest X-ray**; chest CT only if abdominal CT positive or CXR abnormal
- **Non-seminoma** (incl. mixed, **and seminoma histology with elevated AFP**): **chest**/abdominal/pelvic CT up front
- Both: brain MRI if clinically indicated · **sperm banking**
- Nodal spread is predictable — right-sided to inter-aortocaval, left-sided to para-aortic

HISTOLOGY → MARKER PATTERN

Subtype	Markers
Seminoma	AFP always normal ; hCG ↑ in ~15–20%
Embryonal carcinoma	AFP and/or hCG; commonest mixed component
Yolk sac tumor	AFP — the classic AFP producer
Choriocarcinoma	hCG , often >50,000; hemorrhage, brain mets
Teratoma	neither — marker-negative, chemoresistant
Mixed germ cell tumor	reflects whichever components are present

Seminoma hCG comes from **syncytiotrophoblasts** and is compatible with pure seminoma — **only AFP reclassifies**. Very high hCG in choriocarcinoma can cause **hyperthyroidism** (TSH receptor cross-reactivity) and gynecomastia. LDH is non-specific and tracks tumor bulk. Teratoma is marker-negative and chemoresistant — the reason post-chemotherapy masses must be resected.

SERUM TUMOR MARKERS (S)

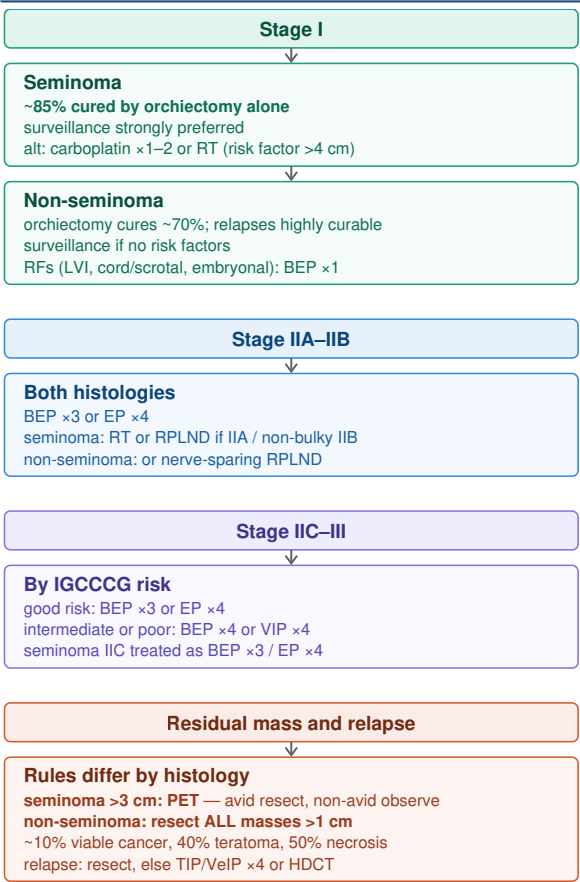
S	LDH	hCG	AFP
S1	<1.5× N	<5,000	<1,000
S2	1.5–10× N	5,000–50,000	1,000–10,000
S3	>10× N	>50,000	>10,000

hCG mIU/mL, AFP ng/mL. pT1a/pT1b (<3 vs ≥3 cm) applies to **pure seminoma only**. cN1 ≤2 cm · cN2 >2–5 cm · cN3 >5 cm. M1a nodal/pulmonary · **M1b non-pulmonary visceral**.

IGCCCG RISK — THE ASYMMETRY

Non-seminoma	Good : testis/retroperitoneal primary, no non-pulmonary visceral mets, AFP<1,000 + hCG<5,000 + LDH<1.5×. Intermediate : markers in the middle band. Poor : mediastinal primary, non-pulmonary visceral mets, or AFP>10,000 / hCG>50,000 / LDH>10×
Seminoma	Good : any primary, no non-pulmonary visceral mets, normal AFP. Intermediate : non-pulmonary visceral mets. There is no poor-risk category — only AFP matters, not hCG or LDH

TREATMENT PATHWAY



REGIMENS & DOSING

Regimen	Dosing	Cycle
BEP (Einhorn)	cisplatin 20/day d1–5 · etoposide 100/day d1–5 · bleomycin 30 units d1, 8, 15 · filgrastim d6–12	q21d
EP	etoposide 100/day d1–5 · cisplatin 20/day d1–5 · pegfilgrastim d5→6	q21d
VIP	etoposide 75 · ifosfamide 1200 · cisplatin 20, all d1–5 · mesna · pegfilgrastim d6	q21d
TIP (2L)	paclitaxel 250 d1 · ifosfamide 1500 d2–5 · cisplatin 25 d2–5 · mesna	q21d
VeIP (2L)	vinblastine 0.11 mg/kg d1–2 · ifosfamide 1200 d1–5 · cisplatin 20 d1–5 · mesna	q21d
HDCT	carboplatin 700 + etoposide 750 — days –5, –4, –3 before stem-cell infusion, ×2 cycles	with PBSC rescue
3L options	gemcitabine + paclitaxel ± oxaliplatin · oxaliplatin + irinotecan · oral etoposide 50–100 mg d1–21	
Pembrolizumab	200 mg q3wk (or 400 mg q6wk) — MSI-H, dMMR, or TMB >10	q3wk

All doses mg/m² unless stated. **Growth factor support is built into every first- and second-line regimen**; mesna with every ifosfamide-containing regimen.

PEARLS

- **The AFP rule overrides histology**. Pure seminoma requires seminoma histology *and a normal AFP*; seminoma histology with an elevated AFP is worked up and treated as a non-seminoma. Elevated bHCG is compatible with pure seminoma
- **Dose holds and reductions are discouraged in the curative (1st and 2nd line) setting** — delay only for severe active infection or illness. The one stated exception: hold or reduce cisplatin for creatinine >1.5 (EP)
- **Residual mass rules differ by histology** — PET works for seminoma but not non-seminoma, because teratoma is not FDG-avid yet is chemoresistant and must be resected
- Non-pulmonary visceral metastases define **intermediate-risk seminoma** and **poor-risk non-seminoma**
- RT or RPLND remains an option only for stage IIA and non-bulky IIB **seminoma** (nodes <3 cm)
- Sperm banking before treatment; restage markers after orchiectomy

Biomarkers: testicular cancer is **marker-driven, not biomarker-driven** — AFP, bHCG and LDH determine stage, risk group and response. The only biomarker-selected therapy in the chapter is pembrolizumab for MSI-H, dMMR, or TMB >10.